Branch Coverage Report

Decision (branch) coverage of the automated unit and integration tests

Project	WealthWise – Wealth Management System (Team-02)
Report	Branch Coverage Report
Date	02-11-2026
Build under test	main @ a2c6973 (Sprint 2 + final fixes, PR #77) + testing-week tests
Environment	Linux test machine · Node.js 22.22 · PostgreSQL 16.15 · Next.js 14.2.35 · TypeScript 5.9 · c8 10.1.3 (V8 coverage) · Chromium via Playwright

Branches covered	Files ≥ 85 %	Files < 75 %
85.0%	**22/50**	**5**
1002 of 1178	files	explained below

Scope and method

- **Tool:** c8 10.1.3 (native V8 coverage) with source maps, so results refer to the TypeScript source lines.
- **Measured code:** `src/lib/**` – all business logic (`services`), data access (`db`), validation, authentication helpers and the error funnel. `--all` is used, so files never loaded by a test count as 0 %.
- **Tests run under coverage:** the full unit suite (73 tests) and database integration suite (54 tests), all passed.
- **Not measured here:** Next.js route handlers (`src/app/api`) and React pages – thin wrappers that run only inside the Next.js server. They are validated by the System Validation Report: all 53 API operations exercised, 59/59 system test cases passed.
- **Reproduce:** `npm run test:coverage` (HTML report in `coverage/index.html`).

How to read branch figures: V8 counts every outcome of a decision (`if/else`, `? :`, `&&`, `||`, `??`, default parameters, optional chaining) as a branch. For TypeScript compiled on the fly it also reports one or two branches on line 1 of each module (the module wrapper); these can never be taken by a test, which is why some fully-tested files show 70–80 % branches with 100 % statements.

Branch coverage by file

File	Branches
lib/	**82.5%**
api.ts	83.9% (26/31)
format.ts	77.8% (7/9)
lib/auth/	**73.2%**
jwt.ts	75.0% (18/24)
password.ts	75.0% (12/16)
session.ts	0.0% (0/1)
lib/db/	**86.9%**
advisor.ts	84.6% (22/26)
assets.ts	88.2% (30/34)
credit-score.ts	90.9% (20/22)
dashboard.ts	81.0% (17/21)
documents.ts	87.5% (35/40)
dues.ts	85.7% (12/14)
health.ts	77.8% (7/9)
imports.ts	87.5% (21/24)
loan-products.ts	84.6% (11/13)

File	Branches
loans.ts	88.9% (48/54)
news.ts	82.6% (38/46)
notifications.ts	88.2% (15/17)
postgres.ts	85.7% (18/21)
prepayments.ts	88.4% (38/43)
projection.ts	89.3% (25/28)
receipts.ts	80.0% (12/15)
reports.ts	90.9% (20/22)
rules.ts	90.9% (20/22)
statement.ts	76.9% (10/13)
transactions.ts	92.3% (24/26)
users.ts	86.7% (13/15)
lib/services/	**85.5%**
credit-score.ts	95.7% (44/46)
csv-parser.ts	77.9% (67/86)
emi.ts	85.0% (17/20)
file-store.ts	88.6% (39/44)
fy.ts	92.0% (23/25)
news.ts	84.9% (45/53)
ocr.ts	64.3% (9/14)
pdf-statement.ts	82.5% (33/40)
prepayment.ts	82.8% (24/29)
projection.ts	93.1% (27/29)
receipt-fields.ts	90.9% (30/33)
recommend.ts	84.6% (22/26)
upload.ts	88.9% (16/18)
lib/validation/	**79.8%**
assets.ts	83.3% (10/12)
auth.ts	81.8% (9/11)
imports.ts	77.8% (7/9)
loans.ts	80.0% (8/10)
prepayment.ts	80.0% (8/10)
projection.ts	71.4% (5/7)
receipts.ts	77.8% (7/9)
recommend.ts	71.4% (5/7)
rules.ts	71.4% (5/7)
transactions.ts	80.0% (8/10)
vault.ts	88.2% (15/17)
All files	**85.05%** (1002/1178)

Files below 80 % branch coverage

File	Branches	Uncovered branches – reason
lib/format.ts	77.8% (7/9)	Module-wrapper branches on line 1 only.
lib/auth/jwt.ts	75.0% (18/24)	Uncovered lines build the cookie options used by the login and logout routes – validated by TC-005 (cookie flags) and TC-007 (cookie cleared).

File	Branches	Uncovered branches – reason
`lib/auth/password.ts`	75.0% (12/16)	Uncovered lines are the dummy hash for unknown emails, used by the login route – validated by TC-006.
`lib/auth/session.ts`	0.0% (0/1)	Reads the session cookie through Next.js request context (`next/headers`), which only exists inside a running server. Validated by system tests TC-005, TC-007, TC-008, TC-009, TC-049.
`lib/db/health.ts`	77.8% (7/9)	Module-wrapper branches on line 1 only.
`lib/db/statement.ts`	76.9% (10/13)	Uncovered branches: users without a PAN or without snapshots – both handled by `maskPan` and empty tables (unit-tested in pdf-statement).
`lib/services/csv-parser.ts`	77.9% (67/86)	Uncovered lines: bank auto-detection fallbacks for layouts not in the test files, and the "Dr/Cr" single-amount column branch.
`lib/services/ocr.ts`	64.3% (9/14)	Uncovered branch is the error path for an unreadable image (tesseract failure), which the sample images do not trigger.
`lib/validation/imports.ts`	77.8% (7/9)	Module-wrapper branches on line 1; the row schema is exercised through the import tests.
`lib/validation/projection.ts`	71.4% (5/7)	Only the module-wrapper branches on line 1 are uncovered; every schema rule is tested (validation.test.ts).
`lib/validation/receipts.ts`	77.8% (7/9)	Module-wrapper branch on line 1 and the `saveToVault: "on"` checkbox value.
`lib/validation/recommend.ts`	71.4% (5/7)	Only the module-wrapper branches on line 1 are uncovered.
`lib/validation/rules.ts`	71.4% (5/7)	Only the module-wrapper branches on line 1 are uncovered.

Decision points covered by targeted tests

Decision	Both outcomes tested by
Login: unknown email vs wrong password (same message)	TC-006 (system), security.test.ts
Session token valid / tampered / wrong secret / expired / alg none	security.test.ts, TC-049
Loan rate and tenure limits (0 / 0.01 / 60 / 60.01; 0 / 1 / 480 / 481)	validation.test.ts, TC-016
Loan edit allowed / refused after payment / refused after prepayment	loans-edit.test.ts, TC-018
EMI payment: unpaid / already paid / other user / last installment closes loan	TC-019 – TC-021, TC-050
Prepayment: reduce tenure / reduce EMI / amount ≥ outstanding / funding asset too small	prepayment tests (unit + db), TC-023 – TC-025
Import row: valid / bad date / missing description / debit / credit / duplicate	csv-parser.test.ts, imports.test.ts, TC-029 – TC-032
Rule precedence: own rule / system rule / longest keyword / no match	rules.test.ts, TC-033
Upload: missing / too large / wrong content / allowed	upload.test.ts, file-store.test.ts, TC-031, TC-035, TC-037
Vault storage writable / not writable (503)	vault.test.ts, TC-038
News feed: fresh cache / stale + reachable / stale + down / no cache (samples)	news-refresh.test.ts, news.test.ts, TC-046
Advisor: assigned client / non-client / individual caller	advisor.test.ts, TC-047, SV-03
Error funnel: validation / HttpError / 23505 / 23514 / 23503 / 22xxx / unknown	security.test.ts

Sign-off

Role	Name	Signature / Date
Test Lead	Likith Kumar	
Security & Traceability Lead	Vasu Tripathi	
Architecture Lead	Nishant Nag	
Requirements / UAT Lead	Madhusudhan P	